

Daniele Famà

✉ daniele.fama03@outlook.com |  daniele-fama |  danielefam |  danielefam.github.io

Seeking a 4–6 month End-of-Studies Internship / Master's Thesis starting April 2027 under a French convention de stage

EDUCATION

Télécom Paris, Institut Polytechnique de Paris

2025 – 2027

M.Sc. in Engineering – Double Degree Program

Paris, France

- Year 1 curriculum: Applied Algebra (Cryptography, Quantum Information, Coding Theory)
- Year 2 specialization: Artificial Intelligence

Politecnico di Torino

2024 – 2025

M.Sc. in Data Science and Engineering – Double Degree Program

Turin, Italy

- Selected coursework: Machine Learning, Deep Learning, Big Data Systems

University of Catania

2021 – 2024

B.Sc. in Computer Engineering – 110/110, summa cum laude (Highest Honors)

Catania, Italy

- Thesis: Design of a Third-Order Tow-Thomas Butterworth Filter for Audio-Band Applications

EXPERIENCE

LTCI, Institut Polytechnique de Paris

2026

Research Intern – Energy-Aware Neural Architecture Search

Paris, France

- Engineered an **energy-aware NAS framework** (Python, PyTorch) to search for optimal sub-networks via differentiable structured pruning and empirical hardware regularizers.
- Physically validated **up to 60% real hardware energy reduction** on ResNet-18 deployed on edge devices (Raspberry Pi 5, NVIDIA Jetson) with **zero accuracy loss**.
- Automated power profiling across **2,000+ layer configurations**, capturing physical sensor telemetry on Jetson and Raspberry Pi boards.

PROJECTS

ARDA VSLNet: Efficient Video-Language Localization on Ego4D

2025

Computer Vision & Deep Learning

Politecnico di Torino

- Enhanced VSLNet with FiLM conditioning for natural-language temporal localization in egocentric videos.
- Reduced model size by **88.47%** through knowledge distillation while outperforming a similarly sized baseline by **19.73%**.
- Applied block-wise distillation to reduce parameters by **47.62%** and GFLOPs by **80.6%** while retaining near-teacher accuracy.

Spoken Digit Classification

2025

Machine Learning

Politecnico di Torino

- Built an audio-classification pipeline using spectrogram representations of the Free Spoken Digit Dataset.
- Ranked **1st among more than 100 teams** on the class public leaderboard.

Handwritten Digit Recognition on Casio fx-9860GIII

2026

Embedded ML

Personal Project

- Deployed an INT8 neural network with fixed-point C inference to a resource-constrained graphing calculator.
- Achieved **95.47% accuracy** on MNIST entirely on-device.

SKILLS

Programming : Python, C/C++, MATLAB

ML and Data : PyTorch, TensorFlow, scikit-learn, Spark

Platforms : NVIDIA Jetson, Raspberry Pi, Linux, Git

Languages : Italian (Native), English (C1, Advanced), French (B1, Intermediate)